Department of Computer Science and Engineering

Course Code:CSE430 **Credits:**

Course Name: Digital Signal Processing (DSP) **Prerequisite:CSE111, MAT120**

Lab 03

Elementary Discrete-Time Signals

I. Lab Overview:

Learn MATLAB programming concepts for some elementary discrete-time signals.

II. Elementary discrete-time signals:

Some elementary discrete-time signals are: Unit Sample Sequence Unit Step signal. Unit Ramp Signal. Exponential signal.

III. Lesson Fit:

There is pre-requisite to this lab: CSE111, MAT120. You should have intensive Programming Knowledge and capability of understand Integral Calculus and Differential Equations.

IV. Acceptance and Evaluation

Performed lab tasks will be evaluated by the Lab Instructor (LI)

a. Short viva will be conducted in each Lab or occasionally to examine your work.
b. You may work in groups but be aware that you will be evaluated individually; hence active participation during the Lab work demonstration is recommended.
c. There will be Lab handout after your work you have to handover it to LI

V. Learning Outcome:

After this Lab, the students will be able to:

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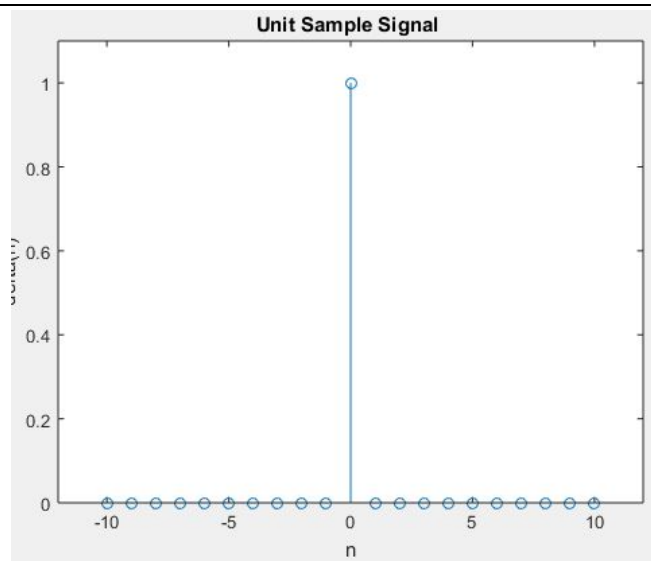
a. Understand basic of elementary discrete-time signals.

VI. Activity Detail

a. Unit Sample Signal (Hour: 1.50) [Details in Lecture Slides]

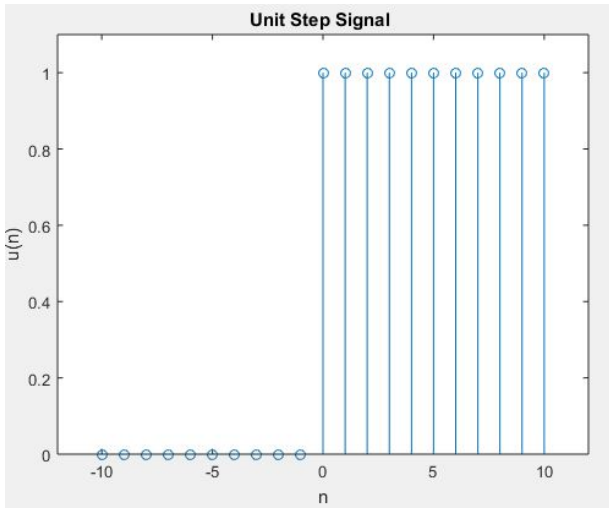
```
close all
clear all
clc

n = -10: 10;
f = (n == 0);
stem(n,f)

xlabel('n');
ylabel('delta(n)');
title('Unit Sample Signal ');
axis([-12 12 0 1.1])
```



b. Unit Step Signal (Hour: 1.50) [Details in Lecture Slides]

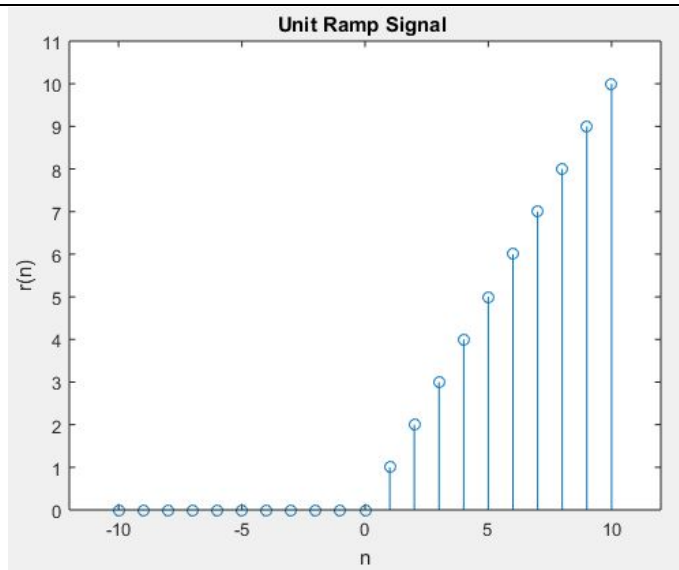
```
close all
clear all
clc
```

```
n = -10: 10;
f = (n >= 0);
stem(n,f)

xlabel('n');
ylabel('u(n)');
title('Unit Step Signal ');
axis([-12 12 0 1.1])
```



c. Unit Ramp Signal (Hour: 1.50) [Details in Lecture Slides]

```
close all
clear all
clc

n = -10: 10;
[a,b] = size(n);
r = zeros(a,b);
for i = 1:b
    if n(i) >= 0
        r(i) = n(i);
    else
        r(i) = 0;
    end
end
stem(n,r)
xlabel('n');
ylabel('r(n)');
title('Unit Ramp Signal ');
axis([-12 12 0 11])
```



d. Real Valued Exponential sequence (Hour: 1.50) [Details in Lecture Slides]

```
close all
clear all
clc

n = 0:100;
x = (0.9).^n;

stem(n,x)

xlabel('n');
ylabel('exp(x)');
title('Real Valued Exponential sequence');
%axis([-12 12 0 11])
```

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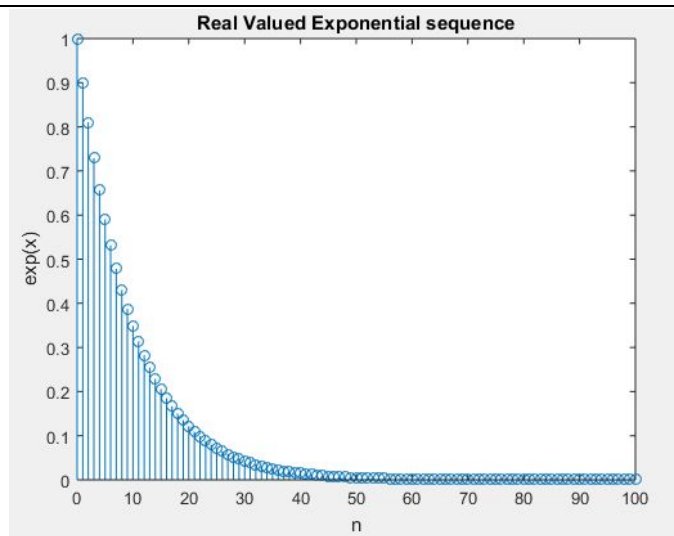


Activity List

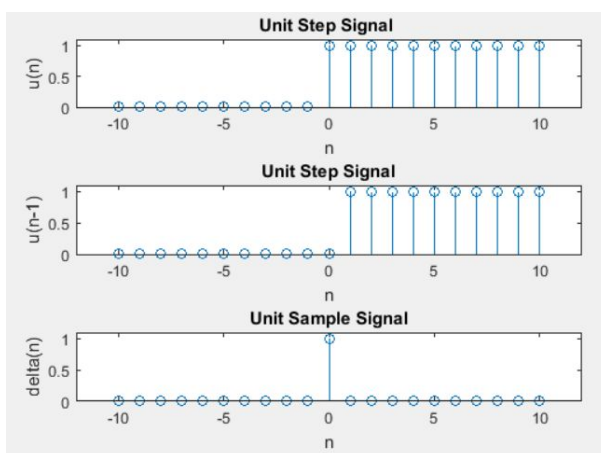
Mark 10 **Time: 1 hour**

Solve the following problems with MATLAB programming language

Question 1. Write a Matlab code to implement the following equation:

$$\delta[n] = u[n] - u[n-1]$$

Sample output:



Question 2. Draw a figure from following complex-valued exponential sequence:

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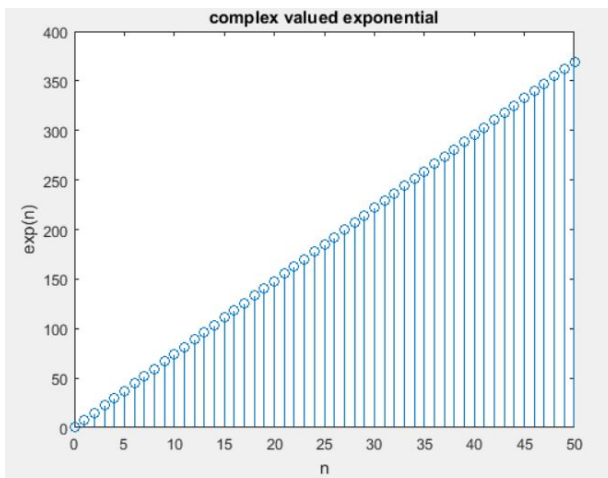
$$x(n) = e^{(\sigma + j\omega_0)n}, \forall n$$

where σ is called an attenuation and ω_0 is the frequency in radians.

Consider σ is 2, ω_0 is 3, and 0>=n>=50.

Sample output



Question 3. Calculate Energy using time series data (i.e., Z001.txt) given in Lab03 folder.

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